

# Def. of Derivative

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx} (x^n) = n \cdot x^{n-1}$$

$$\frac{d}{dx} (\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx} \left( \frac{1}{x} \right) = -\frac{1}{x^2}$$

$$\frac{d}{dx} (e^x) = e^x$$

P.R.  $[A \cdot B]' = A' \cdot B + A \cdot B'$

Q.R.  $\left[ \frac{A}{B} \right]' = \frac{A' \cdot B - A \cdot B'}{B^2}$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\sec x)' = \sec x \cdot \tan x$$

$$(\csc x)' = -\csc x \cdot \cot x$$

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Chain Rule  $[f(g(x))]' = f'(g(x)) \cdot g'(x)$

$$\frac{d}{dx} (\ln x) = \frac{1}{x}$$

$$\frac{d}{dx} (a^x) = a^x \cdot \ln a$$

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\cos^{-1} x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$(\cot^{-1} x)' = \frac{-1}{1+x^2}$$

$$(\sec^{-1} x)' = \frac{1}{x\sqrt{x^2-1}}$$

$$(\csc^{-1} x)' = \frac{-1}{x\sqrt{x^2-1}}$$